

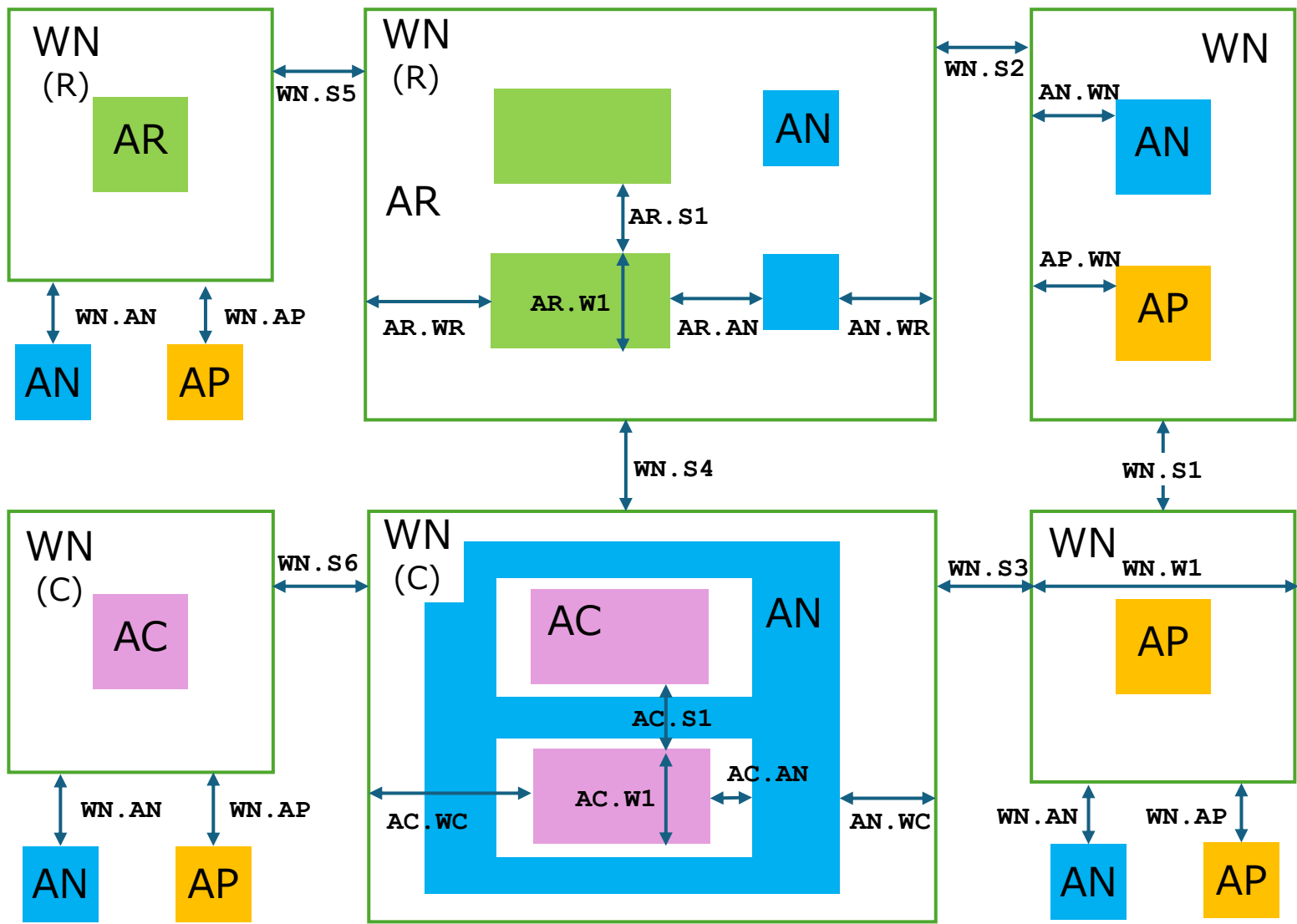
Design Rules

for Drawing Layers

Drawing Layer Table

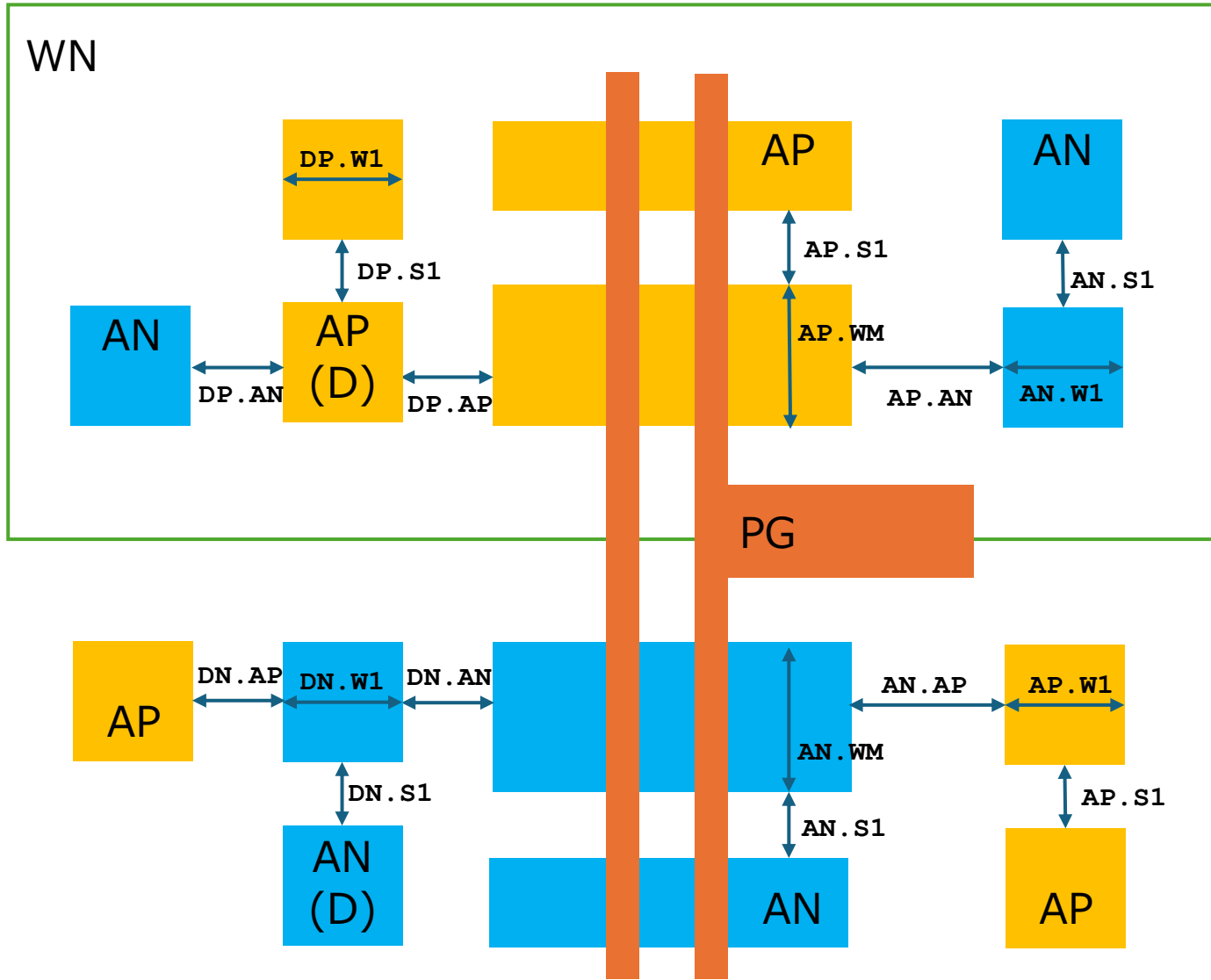
	Original MASK layers				Proposed Drawing layers					
	Name	NUM	TYPE		Name	NUM	TYPE			
1	PSUB	140	0		WN	140	0			
2	NW	36	0					MDP generatioed		
3	HVNW	141	0					MDP generatioed		
4	L	3	0					MDP generatioed		
					AP	3	1	Active for P+		
					AN	3	2	Active for N+		
					AR	3	3	Active for resistor		
					AC	3	4	Active for capacitance		
5	NF	25	0					MDP generatioed		
6	PF	26	0					MDP generatioed		
7	CL	143	0					MDP generatioed		
8	HPBE	144	0					No USE		
9	HNBE	145	0					No USE		
10	PBE	146	0					MDP generatioed		
11	NBE	147	0					MDP generatioed		
12	HPM	33	0					MDP generatioed		
13	RHP	148	0					No USE		
14	SG	8	0							
					PO	8	1	POLY transistor		
					PR	8	2	POLY resistor		
15	PM	35	0					MDP generatioed		
16	NM	7	0					MDP generatioed		
17	R	12	0					MDP generatioed		
18	PSD	9	0					MDP generatioed		
19	NSD	28	0					MDP generatioed		
20	CONT	11	0		CO	11	0			
21	M1	13	0		M1	13	0			
22	TC	19	0		V1	19	0			
23	M2	20	0		M2	20	0			
24	PRO	14	0		PE	14	0			
	TXM1	48	0		TXM1	48	0	M1 Label		
	PIN	48	1		PIN	48	1	M1 Pin		
	TXM2	49	0		TXM2	49	0	M2 Label		
	MASK	63	0		MASK	63	0	Waiver region		
	MEAS	63	1		MEAS	63	1	Measurement region		
	SCRIBE	80	0		SCRIBE	80	0	Waiver region (EDGE SEAL)		
	prBoundary	235	0		prBoundary	235	0	Cell boundary		

WN to AP/AN/AR/AC related



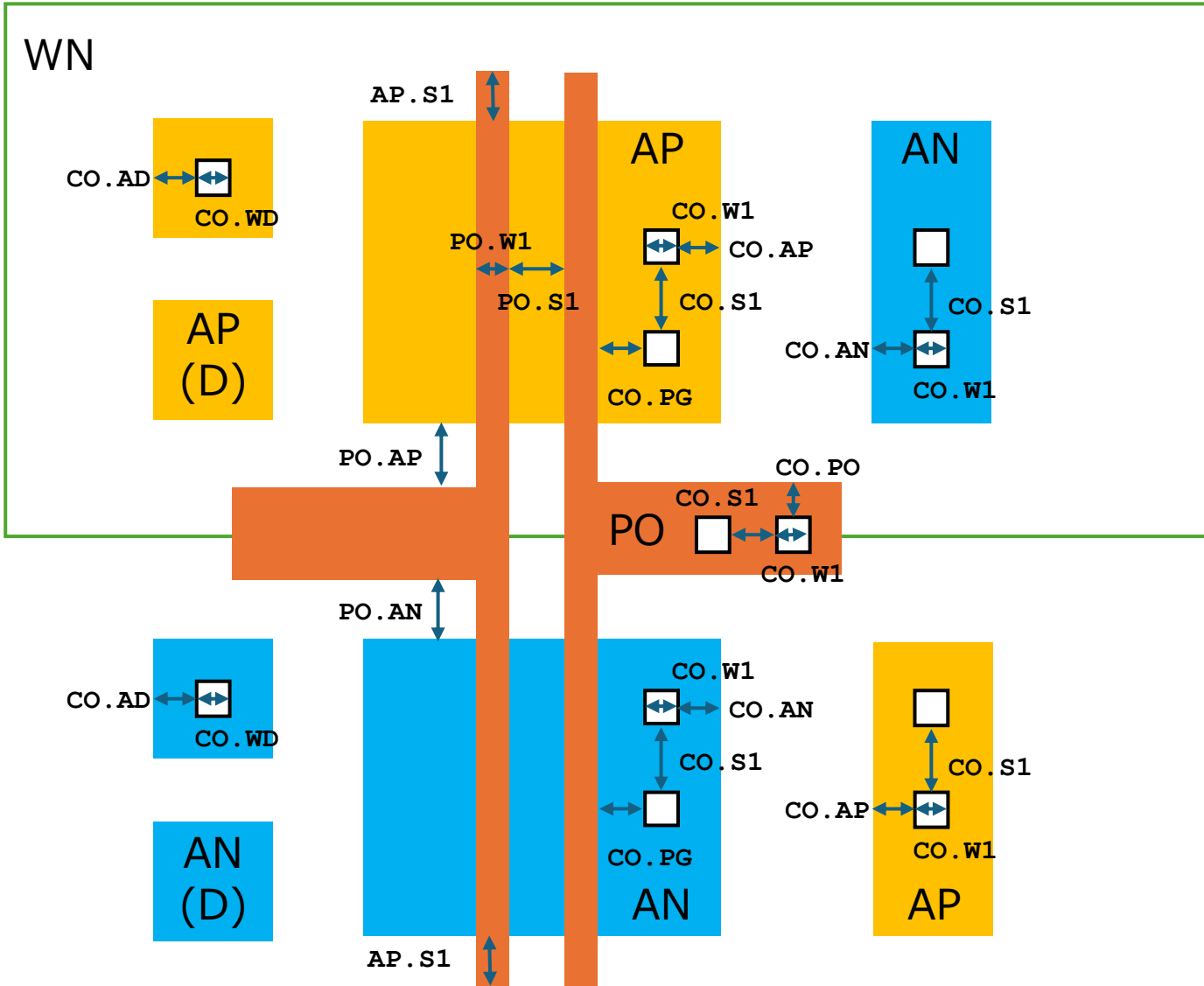
Name	Description	Rule
WN.W1	WN Wmin	8.0
WN.S1	WN-WN Smin	12.0
WN.S2	WN-WN(R) Smin	9.5
WN.S3	WN-WN(C) Smin	12.0
WN.S4	WN(R)-WN(C) Smin	9.5
WN.S5	WN(R)-WN(R) Smin	8.0
WN.S6	WN(C)-WN(C) Smin	12.0
WN.AP	WN-AP Smin	10.0
WN.AN	WN-AN Smin	5.0
AR.W1	AR Wmin	2.8
AR.W2	AR Wmax	20.0
AR.L1	AR Lmin	13.0
AR.L2	AR Lmax	100.0
AR.S1	AR Smin	4.0
AR.WR	AR-WN(R) Emin	10.0
AN.WR	AN-WN(R) Emin	5.0
AR.AN	AR-AN Smin	4.0
AC.W1	AC Wmin	28.5
AC.W2	AC Wmax	120.0
AC.S1	AC Smin	???
AC.WC	AC-WN(C) Emin	???
AN.WC	AN-WN(C) Emin	3.0
AC.AN	AC-AN Smin	2.8
AC.R1	AN must surround AC	

MOS transistor related



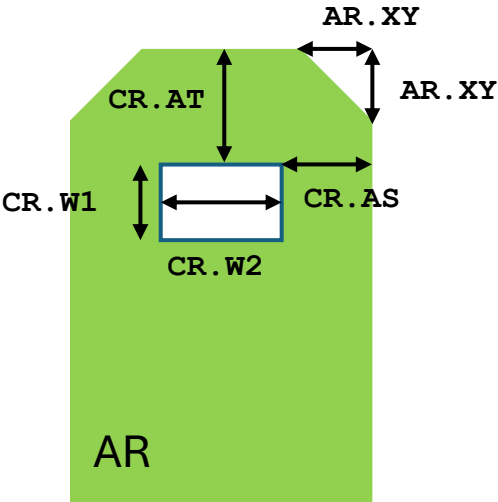
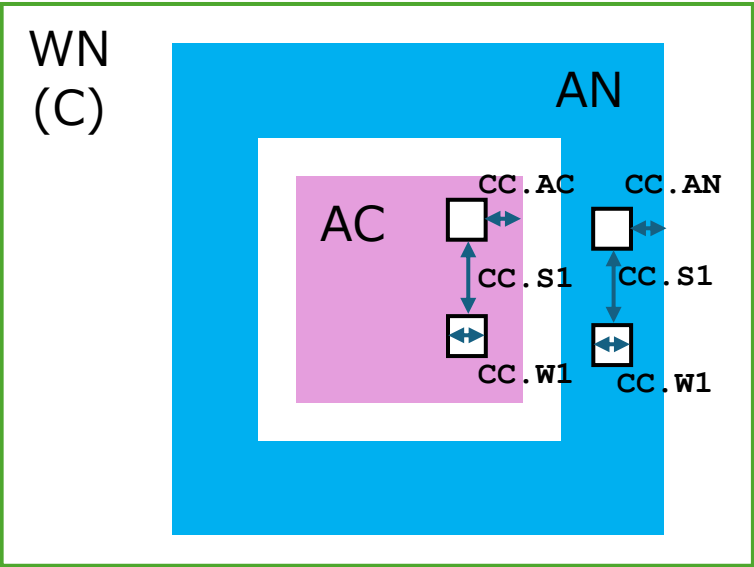
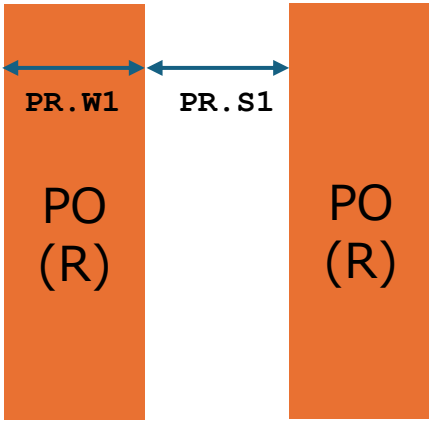
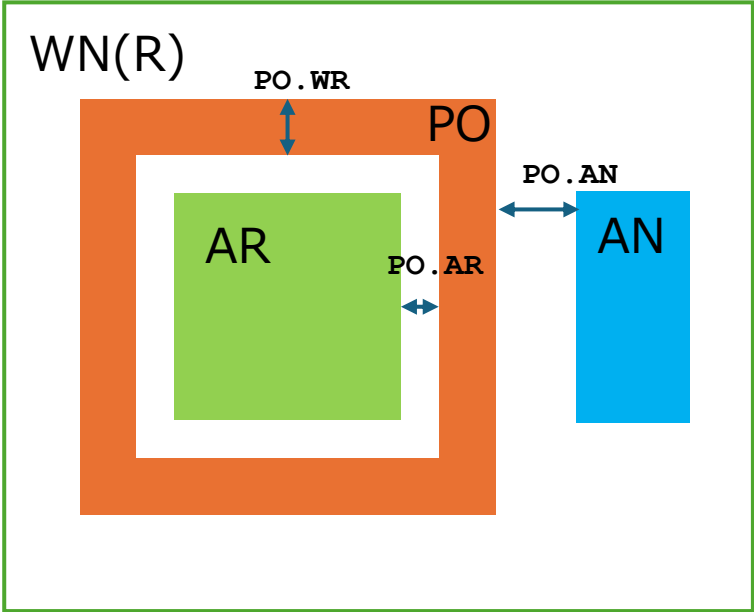
Name	Description	Rule
AP.W1	AP Wmin	1.4
AP.S1	AP Smin	1.4
AP.AN	AP-AN Smin	2.8
AP.WM	PMOS Wmin	3.4
AP.WM	PMOS Wmax	60.0
AP.WN	AP-WN Emin	7.0
DP.W1	AP Wmin	1.4
DP.S1	DP Smin	???
DP.AP	DP-AP Smin	2.8
DP.AN	DP-AN Smin	2.8
AN.W1	AN Wmin	1.4
AN.S1	AN Smin	1.4
AN.AP	AN-AP Smin	2.8
AN.WM	NMOS Wmin	3.4
AN.WM	NMOS Wmax	60.0
AN.WN	AN-WN Emin	5.0
DN.S1	DN Smin	—
DN.AP	DP-AP Smin	2.8
DN.AN	DP-AN Smin	2.8

Contact and Poly related



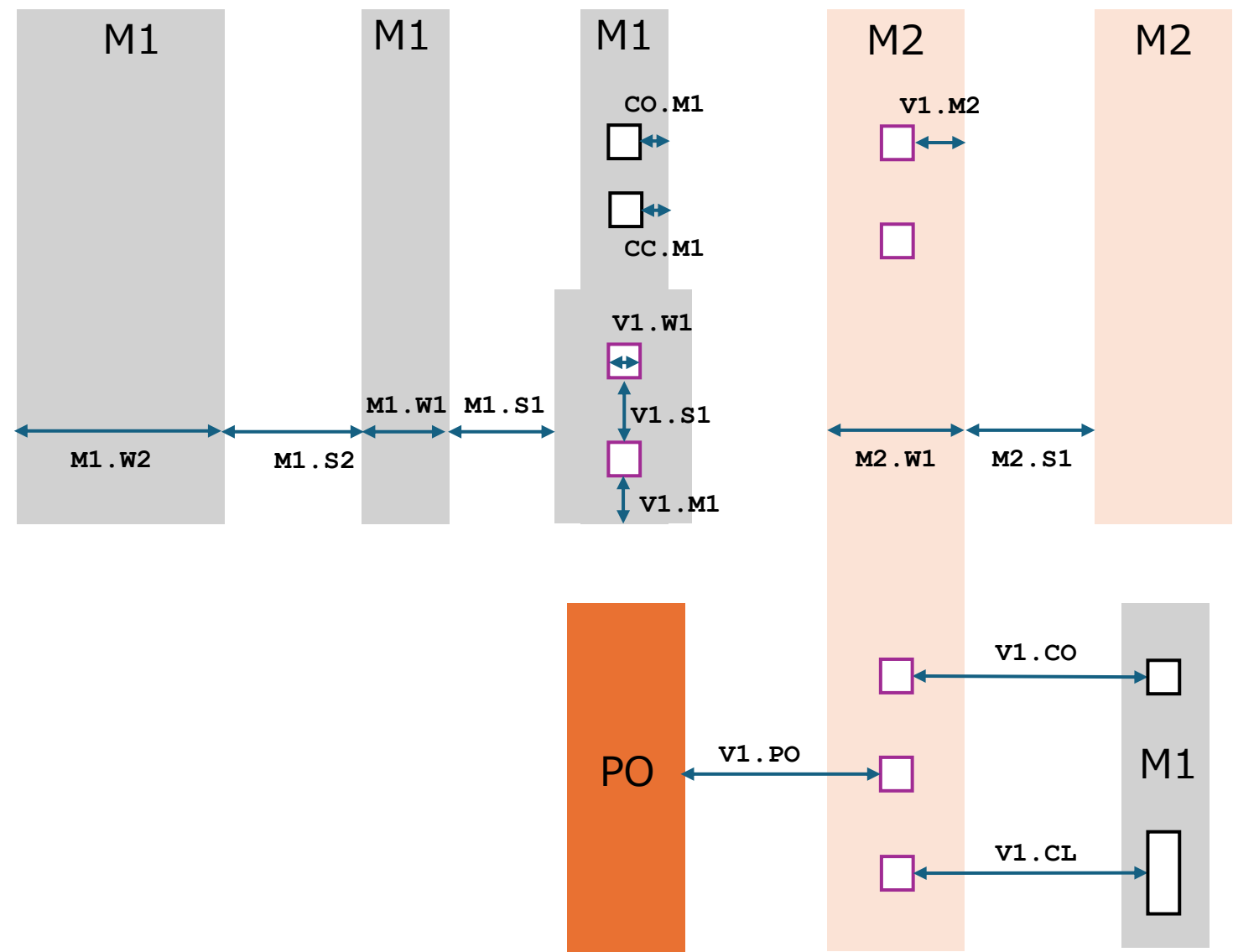
Name	Description	Rule
PO.W1	PO Wmin	1.0
PO.W1	PO Wmax	30.0
PO.S1	PO Smin	1.2
PO.AP	PO-AP Smin	0.4
PO.AN	PO-AN Smin	0.4
PO.EM	MOS Endcap min	1.2
CO.W1	CO Wmin	1.0
CO.S1	CO Smin	1.0
CO.WD	CO(D) Width	1.2
CO.AP	CO-AP Emin	0.8
CO.AN	CO-AN Emin	0.8
CO.PG	CO-PO(G) Smin	1.0
CO.PO	CO-PO Emin	0.8
CO.AD	CO-AD Emin	1.2

Resistor(AR/AS) and Capacitor(CSIO) related



Name	Description	Rule
PO.AR	PO-AR Smin	1.0
PO.WR	PO (AR) Wmin	2.0
PO.AN	PO (AR)-AN Smin	???
PO.R1	PO must surround AR	
PO.R2	PO must tie-down	
PR.W1	PO (R) Wmin	2.8
PR.W1	PO (R) Wmin	4.0
PR.W2	PO (R) Wmax	20.0
PR.L1	PO (R) Lmin	20.0
PR.L2	PO (R) Lmax	100.0
PR.S1	PO (R) Smin	2.0
AR.XT	AR corner cut	1.0
CR.W1	CO (RR) Width	1.0
CR.W2	CO (RR) Wmin	1.0
CR.AT	CO (RR)-AR (Top) Emin	1.5
CR.AS	CO (RR)-AR (Side) Emin	1.0
CC.W1	CO (C) Wmin	1.2
CC.S1	CO (C) Smin	1.2
CC.AC	CO (C)-AC Emin	2.9
CC.AN	CO (C)-AN Emin	1.2
CO.M1	CO-M1 Emin	1.0
CC.M1	CO (C)-M1 Emin	1.2

M1 /V1 /M2 related



Name	Description	Rule
M1.W1	M1 Wmin	1.8
M1.S1	M1 Smin	1.4
M1.W1	M1 (W) Wmin	10.0
M1.S2	M1 (W) Smin	2.0
V1.W1	V1 Windth	1.4
V1.S1	V1 Smin	1.5
V1.M1	V1-M1 Emin	1.0
V1.M2	V1-M2 Emin	1.0
V1.PO	V1-PO Smin	1.2
V1.CO	V1-CO Smin	1.0
V1.CL	V1-CO (L) Smin	1.4
M2.W1	M2 Wmin	3.0
M2.W2	M2 Smin	2.0

質問・確認事項

2025/12/25

DIODE

リファレンス・マニュアルTable I-1-1によればダイオードはP/NどちらもPM+NMインプラと記載

DP : PM+NM

DN : PM+NM

リファレンス・マニュアル図6.1/6.2にも同様の指示があるが、

リファレンス・マニュアル図6.12/6.13に(ESD)では

DP : NM

DN : PM

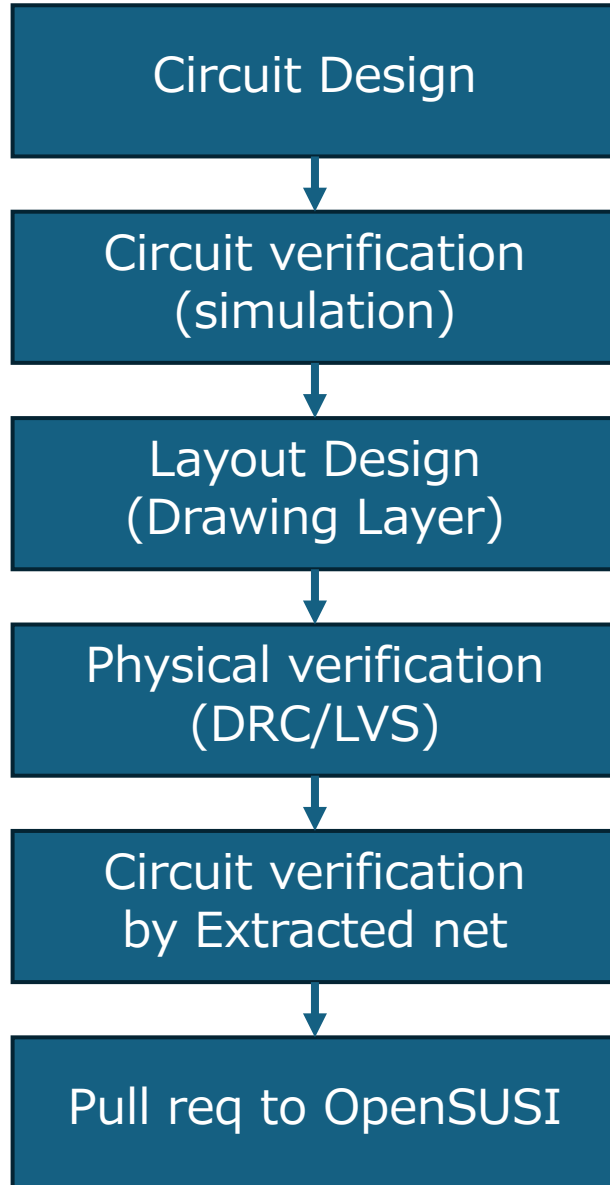
と指示がある。本当はどちらでしょう？違う場合は、Spiceモデルを公開頂けると幸いです。

また、ESD用NMOSのSPICEモデルも公開頂けると幸いです。

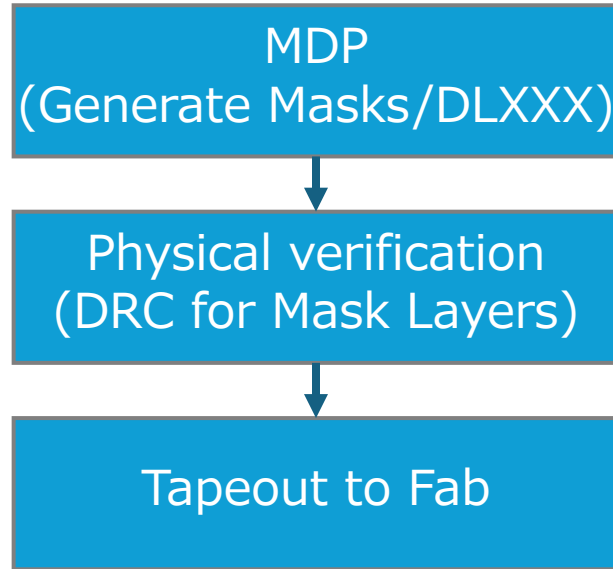
Design to Tapeout Process

Mask Data Preparation

Designer



OpenSUSI



Fab

